API SPECIFICATION DOCUMENT

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Prepared by Pearl Bank to guide technical teams around integration with Wendi E-Wallet APIs

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Document Control

Change History

Date	Version	Description	Author	Reviewed by
24/05/23	1.0	First Draft	Semanda I Moses	James Kasozi
14/03/24	1.1	• Business to Customer (B2C) query. • Business to Customer (B2C) payment. • Business to Business Payment. • E-create. • Transaction Status query.	Okoth Brain	Steven Mbaalu Semanda I Moses James Kasozi
24/11/25	1.1.1	• Group balance • Group information • Group Transfer • Group creation • Group Mini statement	Okoth Brain	Moses Mbabazi Ntare
11/02/26		• Customer to Business (W2B) • Customer to Business query customer details	Okoth Brain	Semanda Isima Moses
10/03/26		• Changed the parameter name in the C2B Api response from wendiReference to thirdPartyReference	Okoth Brain	Semanda Isima Moses

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1 Overview

1.1 Purpose

This document provides the details of the Schema API scheme type provided by the Wendi system, including specific parameters of each service, data type definitions, and status codes. This document is the reference for the third-party to develop their functions based on the Wendi system. Before reading this document, the third-party personnel must first understand the overall design of the API.

1.2 Intended Audience

The intended audiences of this document are:

Third-party Developers

Third-party Testers

Integration Testers

1.3 Using the API

*To use the API, you **MUST** have a xClientID, xClientSecret, Password, and xSignature (this will be formed).*

The data exchange format is Json. The following headers will be required for every request made to the API.

Parameter	Data Type	Description
xClientID	String (Mandatory)	This will be provided.
xClientSecret	String (Mandatory)	This will be provided.
Password	String (Mandatory)	This will be provided.
xSignature	String (Mandatory)	This will be formed as per the details shared in the signature generation section.

1.4 Authentication Security.

Each client shall be provided with the following credentials.

- *xClientID: This will be shared privately to the partner.*
- *xClientSecret: This will be shared privately to the partner.*
- *Password: This will be shared privately to the partner.*
- *xSignature: This will be formed as per the details shared below.*

Before sending a request to the server, the partner will sign the request body with no spaces using the private key. Attach the signature to the request header as xSignature so that the server can verify it.

Verify signatures on the server: On the server, will extract the signature from the request header and verify it using the certificate. If the signature is valid, process the request; otherwise, reject it.

Implementation in Java

```java
import java.nio.charset.StandardCharsets;
import java.security.KeyFactory;
import java.security.PrivateKey;
import java.security.Signature;
import java.security.spec.PKCS8EncodedKeySpec;
import java.util.Base64;
import java.nio.file.Files;
import java.nio.file.Paths;
public class SignatureGenerator {
public static void main(String[] args) throws Exception {
// Read the private key from a file
String privateKeyFilePath = "path/to/your/private-key.pem"; // Replace with the actual file
String pkcs8PrivateKey = new
String(Files.readAllBytes(Paths.get(privateKeyFilePath)), StandardCharsets.UTF_8);
// Original payload even if it has spaces

String payload = "{\"requestId\":\"RDE4526262RQ78922390\",\"msisdn\":\"256704662052\",
\"clientId\":\"526272346\"}";
// Remove spaces from payload for consistent processing
String payloadWithoutSpaces = payload.replaceAll("\\s", "");
byte[] messageBytes = payloadWithoutSpaces.getBytes(StandardCharsets.UTF_8);
// Remove header and footer from the private key, and decode it from Base64
String privateKeyPEM = pkcs8PrivateKey.replace("-----BEGIN PRIVATE KEY-----",
"")
```

```
.replace("-----END PRIVATE KEY-----", "")
.replaceAll("\\s+", "");
byte[] privateKeyDecoded = Base64.getDecoder().decode(privateKeyPEM);
// Generate a private key object from the decoded key
PKCS8EncodedKeySpec keySpec = new PKCS8EncodedKeySpec(privateKeyDecoded);
PrivateKey privateKey = KeyFactory.getInstance("RSA").generatePrivate(keySpec);
// Create a Signature object and initialize it with the private key
Signature sign = Signature.getInstance("SHA256withRSA");
sign.initSign(privateKey);
// Update the Signature object with the message
sign.update(messageBytes);
// Generate the RSA signature
byte[] signatureBytes = sign.sign();
String signatureBase64 = Base64.getEncoder().encodeToString(signatureBytes);
// Print the payloadWithoutSpaces and RSA signature
System.out.println("Payload: " + payloadWithoutSpaces);
System.out.println("RSA Signature: " + signatureBase64);
}
}
```

Implementation in C#

```
    // Read the private key from a file
    string privateKeyPEM = File.ReadAllText(privateKeyFilePath, Encoding.UTF8);

    // Extract key material from PEM format
    string privateKeyHeader = "-----BEGIN RSA PRIVATE KEY-----";
    string privateKeyFooter = "-----END RSA PRIVATE KEY-----";
    int start = privateKeyPEM.IndexOf(privateKeyHeader) + privateKeyHeader.Length;
    int end = privateKeyPEM.IndexOf(privateKeyFooter, start);
    if (start < 0 || end < 0)
    {
        throw new InvalidOperationException("Invalid private key format.");
    }
    string privateKeyBase64 = privateKeyPEM.Substring(start, end - start);
    privateKeyBase64 = privateKeyBase64.Replace("\r", "").Replace("\n", "").Replace(" ", "");
    //privateKeyBase64 = privateKeyBase64.Replace("\r", "").Replace("\n", "");

    // Decode Base64 private key
    byte[] privateKeyDecoded = Convert.FromBase64String(privateKeyBase64);

    // Create RSA object
    RSA rsa = RSA.Create();
    rsa.ImportRSAPrivateKey(privateKeyDecoded, out _);

    byte[] messageBytes = Encoding.UTF8.GetBytes(payload);

    // Sign the data
    byte[] signatureBytes = rsa.SignData(messageBytes, HashAlgorithmName.SHA256, RSASignaturePadding.Pkcs1);

    // Generate the RSA signature
    string signatureBase64 = Convert.ToBase64String(signatureBytes);

    // log the payloadWithoutSpaces and RSA signature
    _logger.LogInformation($"The Payload without whitespaces is: {payload}");
    _logger.LogInformation($"RSA Signature: {signatureBase64}");

    return signatureBase64;
}
catch (Exception ex)
{
    _logger.LogInformation("Signature Generation Exception: " + ex);
    return "failed to generate signature";
}
```

Certificate Requirements

We will require a CA certificate (public key) of a SHA-256 Encryption with RSA-2048 signature A certificate authority (CA) is a trusted organization that issues digital certificates for websites and other entities. You will use your private Key to sign your requests and generate a signature and send it in headers as xSignature.

Self-signed certificates are not allowed. Please generate a certificate under a domain you manage, for example api.yourcompany.com.

1.0. Business to Customer(B2C) query

1.0.1. Customer Lookup Method

This is required when the details of the subscriber are needed. The request body has the parameters as outlined in the table below:

Parameter	Sub parameter	Description	Optional/Mandatory
ClientId		This is the partner Unique identifier	M
Action		Specifies which request is being sent.	M
Data		This a Json object which contains all the data required to complete the request.	M
	requestId	Unique reference from third party application.	M
	msisdn	Customer Wendi account number	M
	clientId	This is the partner Unique identifier	M

Sample Request

Validate Wendi Account Request

```
{
  "clientId": "TEST3PP",
  "action": "ACCT_VALIDATE",
  "Data": {
    "requestId": "RDE4526262RQ789223",
    "msisdn": "256783861192",
  }
}
```

Sample Response

Validate Wendi Account Response

{
  "statusCode": "00",
  "statusDescription": "Process service request successfully.",
  "walletName": "Test User",
  "currency": "UGX",
  "allowCredit": "",
  "allowDebit": ""
}

Endpoint: /wnd/3pp/customer/v1.2/walletaccountlookup

1.1. Business to Customer(B2C) payment

1.1.1. Customer top-up Method.

This is mandatory and is responsible for transaction processing. The request body has the parameters as outlined in the table below:

Parameter	Sub parameter	Description	Optional/Mandatory
clientId		This is the partner Unique identifier	M
action		Specifies which request is being sent.	M
Data		This a Json object which contains all the data required to complete the request.	M
	requestId	Unique reference from third party application.	M
	msisdn	Customer Wendi account number	M
	clientId	This is the partner Unique identifier	M
	narrative		M
	beneficiaryAccount		M
	mobileNumber		M
	beneficiaryName		M
	amount		M
	network		M
	destination		M

Sample Request

Deposit to Wendi Account Request:

```
{
  "clientId": "TEST3PP",
  "action": "WND_PAYMENT_CUSTOMER ",
  "Data": {
```

"requestId": "DES3242512344590UR898",
"narrative": "Deposit to Wendi Account",
"beneficiaryAccount": "256783861192",
"beneficiaryName": "Test Test",
"mobileNumber": "256783861192",
"amount": "500",
"network": "THIRDPARTYID",
"destination": "WENDI"
}
}

Sample Response

Deposit to Wendi Account Response:

{
"statusCode": "00",
"statusDescription": "Process service request successfully.",
"requestId": "DES3242512344590UR898",
"wendiReference": "BE4700876V"
}

Endpoint: /wnd/3pp/customer/v1.2/depositToCustomer

1.2. Business to Business (B2B)

1.2.1. Agent Lookup Method

The Wendi Agent must be validated before the payment is made and parameters are as below:

Parameter	Sub parameter	Description	Optional/Mandatory
clientId		This is the partner Unique identifier	M
action		Specifies which request is being sent.	M
Data		This a json object which contains all the data required to complete the request.	M
	requestId	Unique reference from third party application.	M
	agentShortCode	Wendi Agent account number	M

Validate Sample request: Action is WND_ACCT_VALIDATE_AGENT


```
{
  "clientId": "TEST3PP",
  "action": " WND_ACCT_VALIDATE_AGENT ",
  "Data": {
    "requestId":"38991143898",
    " agentShortCode ":"100031",
  }
}
```


Validate Sample Response:


```
{
  "statusCode": "00",
  "statusDescription": "Process service request successfully.",
  "walletName": "Test User",
  "currency": "UGX",
```

```
  "allowCredit": "",
  "allowDebit": ""
}
```

Endpoint: /wnd/3pp/agent/v1.2/agentAccountLookup

1.2.2. Agent TopUp Method

This is mandatory and is responsible for transaction processing. The request body has the parameters as outlined in the table below:

Parameter	Sub parameter	Description	Optional/Mandatory
clientId		This is the partner Unique identifier	M
action		Specifies which request is being sent.	M
Data		This a Json object which contains all the data required to complete the request.	M
	requestId	Unique reference from third party application.	M
	agentShortCode	Wendi Agent account number to be credited.	M
	narration	Informative statement accompanying a transaction i.e. reason for transfer.	M
	amount	Transaction Amount sent to the Wendi Agent Account.	M
	senderName	Full Name of the transaction initiator.	M

	network	Requesting Application	M
	destination	Receiving Application	M

Sample Request


```
{
  "clientId": "TEST3PP",
  "action": "WND_PAYMENT_AGENT_TOPUP ",
  "Data": {
    "requestId": "DES3242512344590UR898",
    "agentShortCode": "100031",
    "amount": "500",
    " senderName ": "Test Test",
    "msisdn ": "25678289299",
    "narration":"Agent Deposit"
    "network": "THIRDPARTYID",
    "destination": "WENDI"
  }
}
```


Sample Response:


```
{
  "statusCode": "00",
  "statusDescription": "Process service request successfully.",
  "requestId": "DES3242512344590UR898",
  "wendiReference": "BE4700876V"
}
```


Endpoint: /wnd/3pp/agent/v1.2/agentTopUp

1.3. Escrow

1.3.1. E-create Method

Before performing this transaction, an account inquiry be performed using the Agent Lookup Method.

This is mandatory and is responsible for transaction processing. The request body has the parameters as outlined in the table below:

Parameter	**Sub parameter**	**Description**	**Optional/Mandatory**
clientId		This is the partner Unique identifier	M
action		Specifies which request is being sent.	M
Data		This a Json object which contains all the data required to complete the request.	M
	requestId	Unique reference from third party application.	M
	agentShortCode	Wendi Agent short code to be prefunded.	M
	narration	Informative statement accompanying a transaction i.e. reason for transfer.	M
	amount	Transaction Amount sent to the Wendi Agent Account.	M
	senderName	Full Name of the transaction initiator.	M
	network	Requesting Application	M
	destination	Receiving Application	M

Sample Request

{
  "clientId": "TEST3PP",
  "action": "WND_PAYMENT_ECREATE ",
  "Data": {
    "requestId": "DES3242512344590UR898",
    "agentShortCode": "100031",
    "amount": "500",
    " senderName ": "Test Test",
    "msisdn ": "25678289299",
    "narration":"Agent Deposit"
    "network": "THIRDPARTYID",
    "destination": "WENDI"
  }
}

Sample Response:

{
  "statusCode": "00",
  "statusDescription": "Process service request successfully.",
  "walletName": "Test Agent",
  "currency": "UGX",
  "allowCredit": "",
  "allowDebit": ""
}

Endpoint: /wnd/3pp/agent/v1.2/eCreate

1.4. Transaction Inquiry

1.4.1. Transaction Inquiry

This is mandatory and gets the transaction status.
The request body has the parameters as outlined in the table below:

Parameter	Sub parameter	Description	Optional/Mandatory
requestId		Original reference from third party application.	M
msisdn		Customer Wendi account number	M
clientId		This is the partner Unique identifier	M

Sample Request


```
{
  "clientId": " THIRDPARTYID ",
  "requestId": "QC526272728"
}
```


Sample Response


```
{
  "statusCode": "00",
  "statusDescription": "Completed Successful",
  "requestId": "9PSC0006547465",
  "wendiReference": "BH62727277"
}
```


Endpoint: /wnd/3pp/customer/v1.2/transactionInquiry

1.5. Liquidation

Account Liquidation Method

The thirdparty is expected to conform to this request format while implementing their liquidation api.

Sample Request


```
{
  "clientId": "TEST3PP",
  "action": "WND_PAYMENT_LIQUIDATION",
  "Data": {
    "wendiReference": " BH62727277",
    "reasonTypeAlias": "Online Withdraw from Holding",
    " sendingIdentityID": "100031",
    "recipient": "1032201003076",
    "amount":"10000",
    "currency": "UGX",
    "sendingTime": "20240117151306"
  }
}
```


Sample Response:


```
{
  "statusCode": "00",
  "statusDescription": "Completed Successful",
  "requestId": "9PSC0006547465",
  "wendiReference": "BH62727277"
}
```


1.6. Group Transfer

This is mandatory and is responsible for transaction processing. The request body has the parameters as outlined in the table below:

Parameter	Sub parameter	Description	Optional/Mandatory
clientId		This is the partner Unique identifier	M

action		Specifies which request is being sent.	M
Data		This a Json object which contains all the data required to complete the request.	M
	requestId	Unique reference from third party application.	M
	groupCode	Wendi Group account number to be debited.	M
	beneficiaryPhoneNumber	Telephone Number to be notified.	M
	beneficiaryName	The Name of the beneficiary whose account to be credited.	M
	amount	The amount equivalent of the given transaction.	M
	narration	This is a reason why the given transaction is performed.	M

Sample Request

```
{
  "clientId": "TESTUSER",
  "action": "WND_G2P_TRANSFER",
  "Data": {
    "requestId": "890036782926733455678",
    "groupCode": "10000202",
    "beneficiaryPhoneNumber": "256700844482",
    "beneficiaryName": "Paul Johnson",
    "amount": "500",
```

```
    "narration": "P2G Transaction"
  }
}
```

Sample Response

```
{
  "statusCode": "00",
  "statusDescription": "Completed Successful",
  "requestId": "9PSC0006547465",
  "wendiReference": "BH62727277",
}
```

1.7. Group Balance

This is mandatory and is responsible for querying the group balance. The request body has the parameters as outlined in the table below:

Parameter	Sub parameter	Description	Optional/Mandatory
clientId		This is the partner Unique identifier	M
action		Specifies which request is being sent.	M
Data		This a Json object which contains all the data required to complete the request.	M
	requestId	Unique reference from third party application.	M
	groupCode	Wendi Group Account Number to whose balance is to be queried.	M

	accountType	The type of a Group Account	M

Sample Request


```
{
  "clientId": "TESTUSER",
  "action": "WND_CHECK_GP_BALANCE",
  "Data": {
    "requestId": "89003ESWESD2322220253525",
    "groupCode": "10000202",
    "accountType": "A"
  }
}
```


Sample Response


```
{
  "statusCode": "00",
  "statusDescription": "Completed Successful",
  "balance": "10000",
  "currency": "UGX",
}
```


1.8. Group Information

This is mandatory and is responsible for querying the group balance. The request body has the parameters as outlined in the table below:

Parameter	**Sub parameter**	**Description**	**Optional/Mandatory**
clientId		This is the partner Unique identifier	M
action		Specifies which request is being sent.	M
Data		This a Json object which contains all the data required	M

		to complete the request.	
	requestId	Unique reference from third party application.	M
	groupCode	Wendi Group Account Number whose details are to be queried.	M

1.9. Group Ministatement

This is mandatory and is responsible for querying the group balance. The request body has the parameters as outlined in the table below:

Parameter	**Sub parameter**	**Description**	**Optional/Mandatory**
clientId		This is the partner Unique identifier	M
action		Specifies which request is being sent.	M
Data		This a Json object which contains all the data required to complete the request.	M
	requestId	Unique reference from third party application.	M
	startDate	This is the start date for a range for which the ministatement is to be queried.	M
	endDate	This is the end date for a range	M

		for which the ministatement is to be queried.	
	groupCode	The group account for which the ministatement is to be queried.	

Sample Request


```
{
  "clientId": "TEST",
  "action": "WND_G2P_MINISTATEMENT",
  "Data": {
    "requestId": "89003678090323901",
    "startDate": "20250624093000",
    "endDate": "20250626093000",
    "groupCode": "10000202"
  }
}
```


Sample Response


```
{
  "statusCode": "00",
  "statusDescription": "Completed Successful",
  "Data": {
    "ReceiptNumber": "BF526F5362",
    "IdentityPublicName": "TEST GROUP",
    "Currency": "UGX",
    "BalanceAmount": "2000",
    "Details": "test details",
    "CompletionTime": "10000202"
  }
}
```

1.10. Group Creation

This is mandatory and is responsible for querying the group balance. The request body has the parameters as outlined in the table below:

Parameter	Sub parameter	Description	Optional/Mandatory
clientId		This is the partner Unique identifier	M
action		Specifies which request is being sent.	M
Data		This a Json object which contains all the data required to complete the request.	M
	requestId	Unique reference from third party application.	M
	groupName	This is the name of the group to be created.	M
	groupLeaderPhoneNumber	This is the Actively subscribed Wendi phone number of the group Leader.	M

Sample Request

```
{
  "clientId": "TESTUSER",
  "action": "WND_CREATE_GP",
```

"Data": {
"requestId": "89003623421423571012",
"groupName": "TEST GROUP",
"groupLeaderPhoneNumber": "256783861192"
}
}

Sample Response

{
"statusCode": "00",
"statusDescription": "Completed Successful",
"groupName": "TEST GROUP",
"groupCode": "1000234",
}

1.11. Customer to Business (Wallet2Bank)

Customer to Business (Wallet2Bank) C2B

The thirdparty is expected to conform to this request format while implementing their Wallet2Bank api.

The request body has the parameters as outlined in the table below:

Parameter	Sub parameter	Description	Optional/Mandatory
clientId		This is our Unique identifier	M
action		Specifies which request is being sent.	M
Data		This a Json object which contains all the data required to complete the request.	M
	requestId	Unique reference from our system.	M
	msisdn	Customer Wendi account number	M
	clientId	This is our Unique identifier	M
	narrative	This is the transaction description / reason.	M
	beneficiaryAccount	This is the bank Account Number receiving the funds.	M
	mobileNumber	This is the mobile Number of the Bank Customer receiving funds.	M
	beneficiaryName	This is the name of the customer receiving the funds.	M
	amount	This is the amount to be credited	M
	network	Requesting Application	M
	destination	Receiving Application	M

Sample Request

Deposit to Bank Account Request:
{
  "clientId": "TEST3PP",
  "action": "WND_PAYMENT_CUSTOMER ",
  "Data": {
    "requestId": "DES3242512344590UR898",
    "narrative": "Deposit to Wendi Account",
    "beneficiaryAccount": "5361776373783",
    "beneficiaryName": "Test Test",
    "mobileNumber": "256783861192",
    "amount": "500",
    "network": "THIRDPARTYID",
    "destination": "WENDI"
  }
}

Sample Response
Deposit to Bank Account Response:
{
  "statusCode": "00",
  "statusDescription": "Process service request successfully.",
  "requestId": "DES3242512344590UR898",
  "thirdPartyReference": "T637829920347733"
}

1.12. Customer to Business (C2B) query customer details.

The third-party is expected to conform to this request format while implementing their Wallet2Bank api.
The request body has the parameters as outlined in the table below:

Parameter	Sub parameter	Description	Optional/Mandatory
ClientId		This is our Unique identifier	M
Action		Specifies which request is being sent.	M
Data		This a Json object which contains all the data required to	M

		complete the request.	
	requestId	Unique reference from third party application.	M
	beneficiaryAccount	Customer Bank account number	M
	clientId	This is the partner Unique identifier	M

Sample Request

Validate Bank Account Request


```
{
  "clientId": "TEST3PP",
  "action": "ACCT_VALIDATE",
  "Data": {
    "requestId": "RDE4526262RQ789223",
    " beneficiaryAccount ": "637389…..",
  }
}
```


Sample Response

Validate Bank Account Response


```
{
  "statusCode": "00",
  "statusDescription": "Process service request successfully.",
  "beneficiaryName": "Test User",
  "currency": "UGX",
  "allowCredit": "",
  "allowDebit": ""
}
```

1.13. Error responses

The error occurs when there is some missing header field.

{
  "httpCode": "401",
  "httpMessage": "Unauthorized",
  "moreInformation": "Missing the Mandatory header (s)."
}

The error occurs when the signature verification fails.

{
  "httpCode": "401",
  "httpMessage": "Unauthorized",
  "moreInformation": "Invalid Credentials"
}

This error occurs when the clientID passed is invalid.

{
  "httpCode": "401",
  "httpMessage": "Unauthorized",
  "moreInformation": "Invalid client ID"
}

This error occurs when the clientID sends a duplicate requestId.

{
  "statusCode": "408",
  "statusDescription": "Duplicate request",
  "requestId": "DES3242512344590UR898",
  "wendiReference": ""
}

1.14. Response Status

Every request made comes back with a response code. Below is the table that summaries some of the possible response codes and their meaning. Note: Not all error codes have been explicitly documented.

Code	Meaning
00	Success
401	Unauthorized
403	Forbidden
400	Bad Request
404	Not Found
500	Internal Server Error
144	Upload Received Successfully
303	General Error